

FreeIPA Server & Client Configuration

Cluster: Virtual HPC Cluster

IPA Server: hpc-master (192.168.100.1)

IPA Clients: compute1, compute2, login, storage

IPA Domain: hpc.local

IPA Realm: HPC.LOCAL

IPA Admin Password: Admin1234

1. What is FreeIPA?

FreeIPA (Free Identity, Policy and Audit) is an integrated security management solution for Linux environments. It provides centralized authentication, authorization, and account management for all nodes in a cluster.

FreeIPA combines several well-known technologies into a single, easy-to-manage system:

Component	Technology	Purpose
Identity	389 Directory Server (LDAP)	Stores users, groups, hosts
Authentication	MIT Kerberos	Secure ticket-based login
DNS	BIND	Name resolution for the domain
Certificate Authority	Dogtag PKI	Issues SSL/TLS certificates
Web UI	Apache	Browser-based management
Policy	SSSD	Applies access control rules

2. Why Use FreeIPA in an HPC Cluster?

In an HPC cluster, users need to log into multiple nodes — login node, compute nodes, and storage. Without centralized identity management, you would need to create and manage user accounts separately on every single node. FreeIPA solves this by:

- **Centralizing user accounts** — create a user once on the IPA server, it is available on all nodes immediately
- **Single Sign-On (SSO)** — users authenticate once via Kerberos and can access all nodes without re-entering passwords
- **Centralized password policy** — enforce password complexity, expiry, and lockout rules from one place
- **Host-based access control** — control which users can log into which nodes
- **Audit trail** — all authentication events are logged centrally

hpc-master (IPA Server)

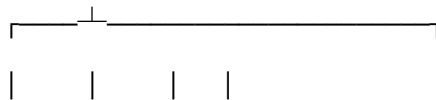
- └─ LDAP Directory — stores users, groups, hosts

- └─ Certificate Auth — manages SSL certificates

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| Domain: hpc.local

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(IPA Client) (IPA Client) (IPA Client) (IPA Client)

Username	Full Name	Password
admin	IPA Administrator	Admin1234
amogh	Amogh User	amogh123
amit	Amit User	amit123

Username	Full Name	Password
admin	IPA Administrator	Admin1234
amogh	Amogh User	amogh123
amit	Amit User	amit123

Note: All IPA server configuration was performed directly on hpc-master via SSH. xCAT was not used for this step.

hostname

Output:

```
[root@hpc-master ~]# hostname -f
hpc-master.hpc.local
[root@hpc-master ~]#
```

4.2 Verify /etc/hosts

cat /etc/hosts | grep hpc-master

Confirmed entry:

192.168.100.1 hpc-master.hpc.local hpc-master

```
hpc-master.hpc.local
[root@hpc-master ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6

127.0.0.1    localhost localhost.localdomain
192.168.100.1 hpc-master.hpc.local hpc-master

192.168.100.11 login.hpc.local    login
192.168.100.12 storage.hpc.local  storage
192.168.100.13 compute1.hpc.local compute1
192.168.100.14 compute2.hpc.local compute2
[root@hpc-master ~]#
```

4.3 Enable IDM Module Stream

The ipa-server package requires the IDM:DL1 module stream to be enabled on Rocky Linux 8:

Check available IDM module streams

dnf module list idm

Enable the DL1 stream

dnf module enable idm:DL1 -y

Install IPA server and DNS packages

dnf install ipa-server ipa-server-dns -y

4.4 Run IPA Server Setup

ipa-server-install \

--domain=hpc.local \

--realm=HPC.LOCAL \

--hostname=hpc-master.hpc.local \

--ip-address=192.168.100.1 \

--no-ntp \

--no-dnssec-validation \

--setup-dns \

--forwarder=8.8.8.8 \

--auto-reverse \

--skip-mem-check

Explanation of installation options:

Option	Meaning
--domain=hpc.local	DNS domain for the IPA installation
--realm=HPC.LOCAL	Kerberos realm name (always uppercase)
--hostname	FQDN of the IPA server
--ip-address	IP address of the IPA server
--no-ntp	Skip NTP configuration — Chrony is already configured
--setup-dns	Configure BIND DNS as part of IPA
--forwarder=8.8.8.8	Upstream DNS forwarder for external queries
--auto-reverse	Automatically create reverse DNS zones
--skip-mem-check	Skip minimum RAM check (required for lab VMs)
--no-dnssec-validation	Disable DNSSEC validation

Interactive prompts answered during setup:

Prompt	Value Entered
Directory Manager password	Admin1234
IPA admin password	Admin1234
NetBIOS domain name	HPC (default, pressed Enter)
Continue with these values?	yes

4.5 Verify IPA Server

Check all IPA services

ipactl status

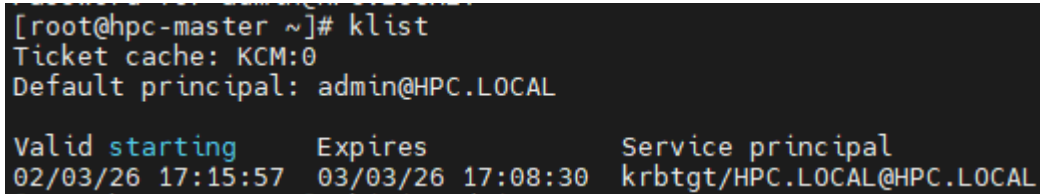
```
The ipa-server-install command was successful
[root@hpc-master ~]# ipactl status
Directory Service: RUNNING
krb5kdc Service: RUNNING
kadmind Service: RUNNING
named Service: RUNNING
httpd Service: RUNNING
ipa-custodia Service: RUNNING
pki-tomcatd Service: RUNNING
ipa-otpd Service: RUNNING
ipa-dnskeysyncd Service: RUNNING
ipa: INFO: The ipactl command was successful
[root@hpc-master ~]# kinit admin
```

Obtain Kerberos ticket for admin

kinit admin

Verify ticket

Klist



```
[root@hpc-master ~]# klist
Ticket cache: KCM:0
Default principal: admin@HPC.LOCAL

Valid starting    Expires          Service principal
02/03/26 17:15:57 03/03/26 17:08:30 krbtgt/HPC.LOCAL@HPC.LOCAL
```

5. Create IPA Users

5.1 Obtain Admin Kerberos Ticket

kinit admin

Enter password: Admin1234

5.2 Create User — amogh

ipa user-add amogh \

--first=Amogh \

--last=User \

--password

Enter password when prompted: amogh123

5.3 Create User — amit

ipa user-add amit \

--first=Amit \

--last=User \

--password

Enter password when prompted: amit123

```
[root@hpc-master ~]# ipa user-add amogh \  
> --first=Amogh \  
> --last=User \  
> --password  
Password:  
Enter Password again to verify:  
-----  
Added user "amogh"  
-----  
User login: amogh  
First name: Amogh  
Last name: User  
Full name: Amogh User  
Display name: Amogh User  
Initials: AU  
Home directory: /home/amogh  
GECOS: Amogh User  
Login shell: /bin/sh  
Principal name: amogh@HPC.LOCAL  
Principal alias: amogh@HPC.LOCAL  
User password expiration: 20260302114645Z  
Email address: amogh@hpc.local  
UID: 543400003  
GID: 543400003  
Password: True  
Member of groups: ipausers  
Kerberos keys available: True
```

```
[root@hpc-master ~]# ipa user-add amit \  
> --first=Amit \  
> --last=User \  
> --password  
Password:  
Enter Password again to verify:  
-----  
Added user "amit"  
-----  
User login: amit  
First name: Amit  
Last name: User  
Full name: Amit User  
Display name: Amit User  
Initials: AU  
Home directory: /home/amit  
GECOS: Amit User  
Login shell: /bin/sh  
Principal name: amit@HPC.LOCAL  
Principal alias: amit@HPC.LOCAL  
User password expiration: 20260302114706Z  
Email address: amit@hpc.local  
UID: 543400004  
GID: 543400004  
Password: True  
Member of groups: ipausers  
Kerberos keys available: True
```

5.4 Verify Users Were Created

List all IPA users

ipa user-find

```
[root@hpc-master ~]# ipa user-find
-----
3 users matched
-----
User login: admin
Last name: Administrator
Home directory: /home/admin
Login shell: /bin/bash
Principal name: admin@HPC.LOCAL
Principal alias: admin@HPC.LOCAL, root@HPC.LOCAL
UID: 543400000
GID: 543400000
Account disabled: False

User login: amit
First name: Amit
Last name: User
Home directory: /home/amit
Login shell: /bin/sh
Principal name: amit@HPC.LOCAL
Principal alias: amit@HPC.LOCAL
Email address: amit@hpc.local
UID: 543400004
GID: 543400004
Account disabled: False

User login: amogh
First name: Amogh
Last name: User
Home directory: /home/amogh
Login shell: /bin/sh
Principal name: amogh@HPC.LOCAL
Principal alias: amogh@HPC.LOCAL
Email address: amogh@hpc.local
UID: 543400003
GID: 543400003
Account disabled: False
```

Check individual users

ipa user-show amogh

ipa user-show amit

```
-----  
[root@hpc-master ~]# ipa user-show amogh  
  User login: amogh  
  First name: Amogh  
  Last name: User  
  Home directory: /home/amogh  
  Login shell: /bin/sh  
  Principal name: amogh@HPC.LOCAL  
  Principal alias: amogh@HPC.LOCAL  
  Email address: amogh@hpc.local  
  UID: 543400003  
  GID: 543400003  
  Account disabled: False  
  Password: True  
  Member of groups: ipausers  
  Kerberos keys available: True  
[root@hpc-master ~]# ipa user-show amit  
  User login: amit  
  First name: Amit  
  Last name: User  
  Home directory: /home/amit  
  Login shell: /bin/sh  
  Principal name: amit@HPC.LOCAL  
  Principal alias: amit@HPC.LOCAL  
  Email address: amit@hpc.local  
  UID: 543400004  
  GID: 543400004  
  Account disabled: False  
  Password: True  
  Member of groups: ipausers  
  Kerberos keys available: True  
[root@hpc-master ~]#
```

6. IPA Client Installation — All Client Nodes

The same steps were applied on all four client nodes — **compute1, compute2, login, and storage** — by SSHing into each node individually from hpc-master.

Note: IPA clients were configured manually via SSH, not using xCAT tools.

6.1 SSH Into Each Client Node

```
ssh root@compute1 # repeat for compute2, login, storage
```

6.2 Install FreeIPA Client Package

```
dnf install freeipa-client -y
```

6.3 Verify /etc/hosts on Each Client

Ensure the IPA server is resolvable on each client:

```
cat /etc/hosts
```

Each client had the following entries:

```
127.0.0.1 localhost localhost.localdomain
```

```
127.0.0.1 compute1.hpc.local compute1 localhost
```

```
127.0.0.1 compute2.hpc.local compute2 localhost
```

```
127.0.0.1 login.hpc.local login localhost
```

```
127.0.0.1 storage.hpc.local storage localhost
```

Important: Ensure hpc-master.hpc.local is resolvable from each client. If DNS is not working, add it manually:

```
echo "192.168.100.1 hpc-master.hpc.local hpc-master" >> /etc/hosts
```

6.4 Run IPA Client Installation

```
ipa-client-install --mkhomedir \  
--server=hpc-master.hpc.local \  
--domain=hpc.local \  
--realm=HPC.LOCAL \  
--hostname=$(hostname -f) \  
-p admin -w Admin1234 \  
--unattended
```

Explanation of client installation options:

Option	Meaning
--mkhomedir	Automatically create home directory on first login
--server	FQDN of the IPA server
--domain	IPA DNS domain
--realm	Kerberos realm
--hostname	FQDN of this client node
-p admin	IPA admin username for enrollment
-w Admin1234	IPA admin password for enrollment
--unattended	Run without interactive prompts

Interactive prompts answered during client setup:

Prompt	Value Entered
Proceed with fixed values and no DNS discovery?	yes
Configure chrony with NTP server?	yes
NTP source server address	(pressed Enter to skip — Chrony already configured)
NTP source pool address	(pressed Enter to skip)
Continue to configure the system with these values?	yes
Password for admin@HPC.LOCAL	Admin1234

7. Verification — IPA Users on Client Nodes

After IPA client installation, verify IPA users are accessible on each node:

Check if IPA user amogh is recognized

id amogh

Expected output:

uid=1234500001(amogh) gid=1234500001(amogh) groups=1234500001(amogh)

Switch to IPA user

su - amogh

Expected: Login succeeds and home directory is created at /home/amogh.

Verify Kerberos authentication works

kinit amogh

Enter password: amogh123

```
[root@compute1 ~]# id amogh
uid=543400003(amogh) gid=543400003(amogh) groups=543400003(amogh)
[root@compute1 ~]# kinit amogh
Password for amogh@HPC.LOCAL:
Password expired. You must change it now.
Enter new password:
Enter it again:
[root@compute1 ~]# kinit amogh
Password for amogh@HPC.LOCAL:
[root@compute1 ~]#
[root@compute1 ~]#
[root@compute1 ~]# su - amogh
Creating home directory for amogh.
[amogh@compute1 ~]$
[amogh@compute1 ~]$
[amogh@compute1 ~]$
[amogh@compute1 ~]$
[amogh@compute1 ~]$ logout
```

8. Common IPA Administration Commands

User Management

Add a new user

```
ipa user-add <username> --first=<First> --last=<Last> --password
```

Delete a user

```
ipa user-del <username>
```

Disable a user account

```
ipa user-disable <username>
```

Enable a user account

```
ipa user-enable <username>
```

Change a user's password

```
ipa passwd <username>
```

List all users

```
ipa user-find
```

Show user details

```
ipa user-show <username>
```

Group Management

Create a group

```
ipa group-add <groupname> --desc="Description"
```

Add user to group

```
ipa group-add-member <groupname> --users=<username>
```

List all groups

ipa group-find

Show group members

ipa group-show <groupname>

Host Management

List all enrolled hosts

ipa host-find

Show host details

ipa host-show <hostname>

Kerberos

Get Kerberos ticket

kinit <username>

List current tickets

klist

Destroy ticket

kdestroy

IPA Service Control

Check status of all IPA services

ipactl status

Start all IPA services

ipactl start

Stop all IPA services

ipactl stop

Restart all IPA services

ipactl restart
